5-state estimator without a': kinematic gain model, every a'-free defect repaired

2026-09-20 — code: `motion_control_law_eq17_5state_noap.m`; algebra check: `check_5state_noap_algebra.py`

0. What this layer is

The gain a_x is an unknown, slowly varying parameter. Nothing is assumed about *why* it varies: no wall function $c(\bar{h})$, no slope a', no use of the commanded step $\Delta\bar{w}_d$ in the gain model. The only statement about a_x is that it is locally a straight line in time. Everything else in the estimator is the honest description of what the loop actually does (timing, readout statistics, noise paths). The consequences of the missing knowledge are stated in §10 as testable predictions, not as tuning.

1. Setting and timing (per axis, $d = 2$)

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\}, \qquad \delta x[k] = x_d[k] - x[k], \qquad \delta x_m[k] = \delta x[k-d] + n_x[k].$$

Thermal kick $t[k] := a_x[k] f_T[k]$, white, $\mathrm{Var}\, t[k] = 4k_B T\, a_x[k]$. Sensor noise n_x white, σ_n^2.

Call k of the controller, in this order: (i) predict from time $k-1$ to k; (ii) update with $y[k]$; (iii) compute $f_{dx}[k]$ from the posterior $\hat{x}[k|k]$. The force $f_{dx}[k]$ acts over $[k, k+1]$. Hence the transition $k-1 \to k$ is driven by $f_{dx}[k-1]$ (item B), and the law consumes $\hat{a}_x[k|k]$, not the previous posterior (item C).

2. Control law

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k|k]\Big\{\Delta x_d^d[k] + \alpha\Big[\delta x_m[k] - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i|k-i]\, f_{dx}[k-i]\Big]\Big\}, \quad \alpha := 1 - \lambda_c}$$

$\Delta x_d^d[k] = x_d[k+1] - \lambda_c\, x_d[k] - \alpha\, x_d[k-d]$ is the command (used as in every sibling; it is not the step-size feed-forward of the 4-state line, which enters the *gain* model and is absent here).

3. Gain model (the whole content of the layer)

$$a_x[k+1] = a_x[k] + \delta a_x[k], \qquad \delta a_x[k+1] = \delta a_x[k].$$

- $Q_{44} = 0$. In the Vpersonal sibling $Q_{44} = \mathrm{Var}(\delta a_{ram}) = \frac{2}{1+\lambda_c}(aK_h/R)^2\sigma_{\delta h}^2$ is the amplitude of $a' x_{ram}$; without a' it cannot be booked. Unbooked size: stationary $\mathrm{Var}(a_{ram})/a^2 \approx 1\%$ near the wall (measured 0.94% at $\bar{h} = 1.56$, 2026-06-10).
- $Q_{55} = 0$. A non-zero value is a prior on a'' or a' and therefore outside this layer. The single prior number of the layer is $\sqrt{P_{55}[0]}$ (§9).
- The $\delta x_{\delta af}^d$ term of the sibling's ε (past a_{ram} through past forces) and the $\sum \mathrm{Var}(\delta a_{ram})$ delay term of R_{22} are the same quantity and are dropped with it (both measured negligible on 2026-06-08 even when doubled).

4. Closed loop with exact past-error bookkeeping (item C)

Let $e_a[k] = a_x[k] - \hat{a}_x[k|k]$, $e_{\delta a}[k] = \delta a_x[k] - \delta\hat{a}_x[k|k]$. Under the model $a_x[k-i] = a_x[k] - i\,\delta a_x[k]$, and with the law having used $\hat{a}_x[k-i|k-i]$ at step $k-i$,

$$a_x[k-i] - \hat{a}_x[k-i|k-i] = e_a[k] - i\,e_{\delta a}[k] + c_i[k], \qquad c_i[k] := \big(\hat{a}_x[k|k] - \hat{a}_x[k-i|k-i]\big) - i\,\delta\hat{a}_x[k|k].$$

$c_i[k]$ is the sum of the update corrections the filter applied to $\hat{a}_x$ over the last i steps (zero if no update moved it); it is known at call k after the update. The sibling's approximation A3 ("past gain errors ignore the L corrections") is replaced by this identity. Substituting into $a_x f_{dx} - a_x\bar{\bar{f}}_{dx}$ (5state_derivation, "Motion error"; $\delta x_D \equiv 0$):

$$\boxed{\begin{aligned} \delta x[k+1] &= \lambda_c\,\delta x[k] - F_{dx}[k]\,e_a[k] + dF_{dx}[k]\,e_{\delta a}[k] - u_c[k] - \alpha\,n_x[k] - \varepsilon_T[k], \\ F_{dx}[k] &= f_{dx}[k] + \alpha\textstyle\sum_{i=1}^{d} f_{dx}[k-i], \qquad dF_{dx}[k] = \alpha\sum_{i=1}^{d} i\,f_{dx}[k-i], \\ u_c[k] &= \alpha\textstyle\sum_{i=1}^{d} f_{dx}[k-i]\,c_i[k], \qquad \varepsilon_T[k] = t[k] + \alpha\,t[k-1] + \alpha\,t[k-2]. \end{aligned}}$$

Checked symbolically (C1 of the script): the identity is exact, no residual.

5. Thermal memory: MA(2) augmentation (item E)

ε_T shares $t[k-1], t[k-2]$ with $\varepsilon_T[k+1], \varepsilon_T[k+2]$; a white Q_{33} mis-states its DC power by $(1+2\alpha)^2/(1+2\alpha^2) = 2.17$ at $\lambda_c = 0.7$. Augment $m_1[k] := t[k-1]$, $m_2[k] := t[k-2]$:

$$\delta x[k+1] \ni -t[k] - \alpha m_1[k] - \alpha m_2[k], \qquad m_1[k+1] = t[k], \qquad m_2[k+1] = m_1[k].$$

The only process noise is $t[k]$, entering δx_3 with -1 and m_1 with $+1$: $\mathbf{q} = g\,t[k]$, $g = [0\,0\,{-1}\,0\,0\,1\,0]^\top$, $\mathbf{Q} = 4k_BT\,\hat{a}_x\,gg^\top$ (rank 1; C3 of the script).

6. Sensor noise double path as a correlated-noise KF (item D)

$n_x[k]$ sits in $r_1[k]$ and in $q_3[k]$ ($-\alpha n_x[k]$): $S_{31} = -\alpha\sigma_n^2$, set to zero in every sibling. Exact treatment: $n_x[k] = y_1[k] - \delta x_1[k]$, so

$$-\alpha\,n_x[k] = +\alpha\,\delta x_1[k] - \alpha\,y_1[k].$$

The first term is a state coupling $\tilde{F}_e(3,1) = +\alpha$, the second a known input. In the estimator (C2 of the script)

$$\hat{x}_3^-[k+1] \ni \lambda_c\,\hat{x}_3[k|k] - \alpha\,\big(y_1[k] - \hat{x}_1[k|k]\big) = \lambda_c\,\hat{x}_3[k|k] - \alpha\,r_1^+[k],$$

r_1^+ = posterior residual of y_1 (the filter's estimate of $n_x[k]$). n_x leaves $\mathbf{Q}$ entirely; its remaining uncertainty is carried by P_{11} through $\tilde{F}_e(3,1)$. This is `formC_b`'s `nw_mcorr` ($\tilde{F} = F - MR_1^{-1}H_1$, $\tilde{Q} = Q - MM^\top/R_1$, $M_3 = -\alpha\sigma_n^2$).

7. Readout y_2: whitening and colour (items A, G)

$a_{xm}[k] = (1-a_{cov})\,a_{xm}[k-1] + a_{cov}\,u[k]$, $u[k] = (dx_r^2[k] - C_n\sigma_n^2)/(C_{dpmr}\,4k_BT)$, $\mathbb{E}\,u[k] = a_x[k-d]$. Feeding a_{xm} every step over-counts its information by $(2-a_{cov})/a_{cov} = 39$ and adds an EWMA lag of $(1-a_{cov})/a_{cov} = 19$ steps that the $-d$ in H_2 does not cover (for a gain rate δa that is a bias $19\,\delta a$). Both vanish with

$$\boxed{\begin{aligned} &y_2'[k] = a_{xm}[k] - (1-a_{cov})\,a_{xm}[k-1] = a_{cov}\,u[k], \qquad H_2' = a_{cov}\,[0\;0\;0\;1\;{-d}\;0\;0], \\ &R_{22}' = a_{cov}^2\,\mathrm{Var}(u) = 2\,a_{cov}^2\,\mathrm{IF}_{white}\,(\hat{a}_x + \xi)^2, \qquad \xi = \tfrac{C_n}{C_{dpmr}}\tfrac{\sigma_n^2}{4k_BT}, \end{aligned}}$$

Isserlis $\mathrm{Var}(dx_r^2) = 2\mathrm{Var}(dx_r)^2$; $\mathrm{IF}_{white} = 1 + 2\sum_{\tau\geq 1}\rho_u(\tau)$ is the $s = 1$ colour sum of the builder (`IF_abc_white`, 08-27), the raw-readout IF_{eff} being the $s = 1 - a_{cov}$ sum and wrong here. Rule: $H_2 \propto a_{cov}$, $R_{22} \propto a_{cov}^2$, $K_2 \cdot \mathrm{innov}$ independent of a_{cov} — the acceptance test of §10. Gates as in the sibling (G1 warm-up, G2 readout level, G3 $\bar{h}_{safe}$); whitening seeded with $a_{xm}[-1] = \hat{a}_x[0]$ (prefill).

8. Estimator

State $\mathbf{x} = [\delta x_1\ \delta x_2\ \delta x_3\ a_x\ \delta a_x\ m_1\ m_2]^\top$, $\delta x_1 = \delta x[k-2]$. Predict at call k ($k-1 \to k$), $\hat{\mathbf{x}} = \hat{\mathbf{x}}[k-1|k-1]$:

$\hat{x}_1^- = \hat{x}_2$, $\hat{x}_2^- = \hat{x}_3$, $\hat{x}_3^- = \lambda_c\hat{x}_3 - \alpha\hat{x}_6 - \alpha\hat{x}_7 - \alpha\, r_1^+[k-1] - u_c[k-1]$, $\hat{x}_4^- = \hat{x}_4 + \hat{x}_5$, $\hat{x}_5^- = \hat{x}_5$, $\hat{x}_6^- = 0$, $\hat{x}_7^- = \hat{x}$

$$\tilde{\mathbf{F}}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \alpha & 0 & \lambda_c & -F_{dx}[k-1] & dF_{dx}[k-1] & -\alpha & -\alpha \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \qquad \mathbf{Q}[k] = 4k_BT\,\hat{a}_x[k-1|k-1]\,gg^\top, \qquad P^- = \tilde{\mathbf{F}}_e P\tilde{}$$

$F_{dx}[k-1] = f_{dx}[k-1] + \alpha(f_{dx}[k-2] + f_{dx}[k-3])$, $dF_{dx}[k-1] = \alpha(f_{dx}[k-2] + 2f_{dx}[k-3])$; $u_c[k-1] = \alpha\sum_{i=1}^{2} f_{dx}[k-1-i]\,c_i[k-1]$ with the posterior buffers $\hat{a}_x[k-1|k-1], \hat{a}_x[k-2|k-2], \hat{a}_x[k-3|k-3]$. The realised force is kept as regressor (Meng ledger 35/36: with the stale-consumption input removed the endogeneity residual is $\leq 2\%$; no `fdet`).

Update at call k: $\mathbf{y} = [\delta x_m[k];\ y_2'[k]]$, $\mathbf{H} = [1\,0\,0\,0\,0\,0\,0;\ H_2']$, $\mathbf{R} = \mathrm{diag}(\sigma_n^2, R_{22}')$, Joseph form. Then §2 with $\hat{a}_x[k|k]$; store $r_1^+[k] = \delta x_m[k] - \hat{x}_1[k|k]$ and c_i buffers for the next call.

9. Initialisation (item F)

Seed $\hat{a}_x[0] = a_{nom}$ (far-field physics, $c \to 1$; on the trajectories with cheat init $\hat{a}_x[0] = a_x(\bar{h}_0)$), $\delta\hat{a}_x[0] = 0$, $\hat{m} = 0$. $P[0] = \mathrm{blkdiag}\big(P_5^{DARE},\ P_{44}[0],\ P_{55}[0]\big)$ mapped to slots $[1\,2\,3\,6\,7]$, 4, 5: the position/memory block is the y_1-only DARE of $\tilde{F}_5 = \tilde{F}_e(\{1,2,3,6,7\})$ at $F_{dx} = 0$ with $\mathbf{Q}_5 = 4k_BT\hat{a}_x[0]\,g_5g_5^\top$ (a fixed point exists: δx_3 is driven); the gain block is a prior, not a DARE (a noise-free integrator has no fixed point, 07-17).

$\sqrt{P_{44}[0]} = 0.5\,a_{nom}$ (two-sided, covers $a_{nom}/c_\perp$ down to $\bar{h} \approx 1.2$), $\sqrt{P_{55}[0]} = a_{nom}/400$ per step (a full chang

$\sqrt{P_{55}[0]}$ is the only number in the layer that encodes "how fast the gain may move"; it acts on the first learning window only, since $Q_{55} = 0$ lets P_{55} shrink monotonically afterwards.

10. Acceptance (pre-registered) and predictions

1. a_{cov} invariance: $a_{cov} \in \{0.025, 0.05, 0.1\}$ leaves $\hat{a}_x$ unchanged to floating precision except at G2 gate steps (the whitening fingerprint, 08-27 step 4).
2. y_1 honesty: $\mathrm{Var}(\mathrm{innov}_1)/\bar{S}_{11} \approx 1$ in the hold (stale-consumption and n_x path both booked; Meng ledger 25 had 2.5 before the fix).
3. P honesty: across-seed $\mathrm{sd}(\hat{a}_x - a_x)$ vs $\sqrt{P_{44}}$ in the hold windows, ratio in $[0.7, 1.4]$.
4. Prediction of the missing knowledge (not a fault of the estimator, the definition of the layer): P_{44}, P_{55} shrink monotonically, so the estimate tends to the least-squares straight line through

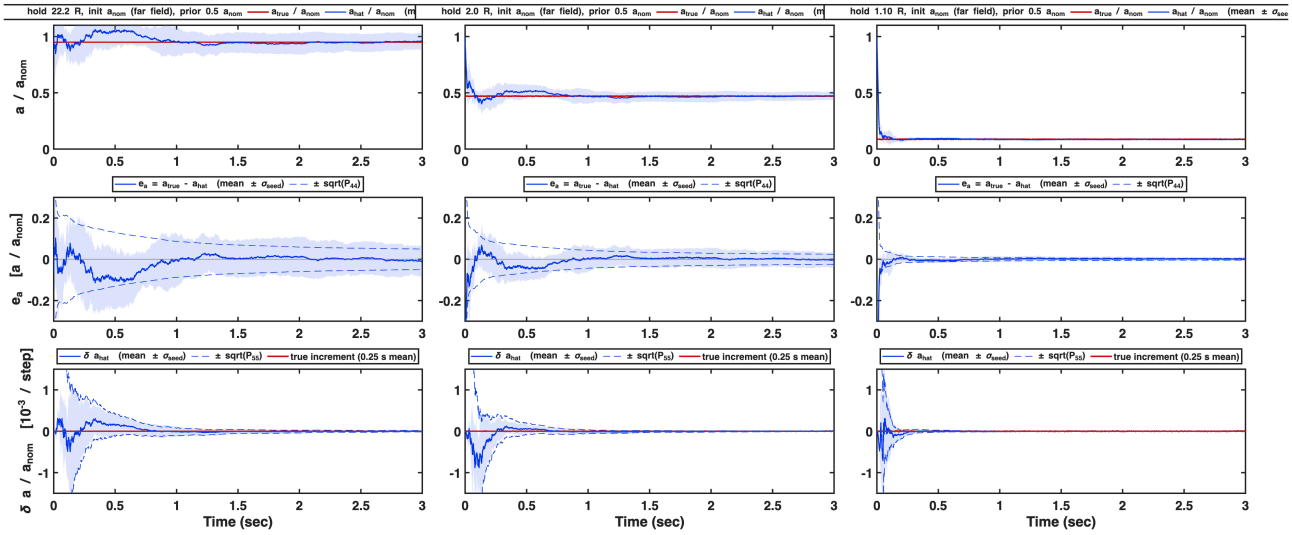
the whole y_2' history. On the ramp (single direction) the line is right; on the 1 Hz oscillation the true rate reverses every half period, the frozen rate cannot follow, and e_a grows from each turning point and is pulled back at the next — a triangular error with amplitude of order the gain swing itself. The 4-state line's $a'\,\Delta\bar{w}_d$ is the answer to exactly this, and nothing in this layer can substitute for it without a prior on a'.

11. Unmodelled, listed

Readout follows the applied gain (loop-pole shift, formB $S \approx 0.32$); residual MA(2) of the whitened u ($\rho \approx [0.6, 0.17]$ at $\lambda_c = 0.4$); a_{pd} mean-centering residual during motion; a_{ram} (1%, §3); cross-axis correlation of the thermal kick (moot with $Q_{44} = 0$).

12. Battery (2026-09-20, plane plant, 10 seeds, z axis; `run_5state_noap_battery.m` / `plot_5state_no`

Acceptance 1 PASS: $a_{cov} \in \{0.025, 0.1\}$ moves $\hat{a}_x$ by $\leq 8 \times 10^{-15}\, a_{nom}$ (raw-readout arm: 0.16–0.24). Acceptance 2: y_1 honesty 0.97 / 0.96 / 0.82 in hold0 / descent / hold-end, **1.76 in the oscillation**; LOO signatures: `ma2_aug` off 2.37, old ordering 1.99, `corr_input` off identical (its magnitude is $\sim 10^{-4}$ nm here: the law reads δx_m raw, so only $\hat{a}_x$ is consumed stale). Acceptance 3: hold P honesty 1.53 / 1.53 / 1.76 at 22.2 / 2.0 / 1.10 R — outside [0.7, 1.4]; see `probe_5state_noap_honesty.m` for the discriminators.

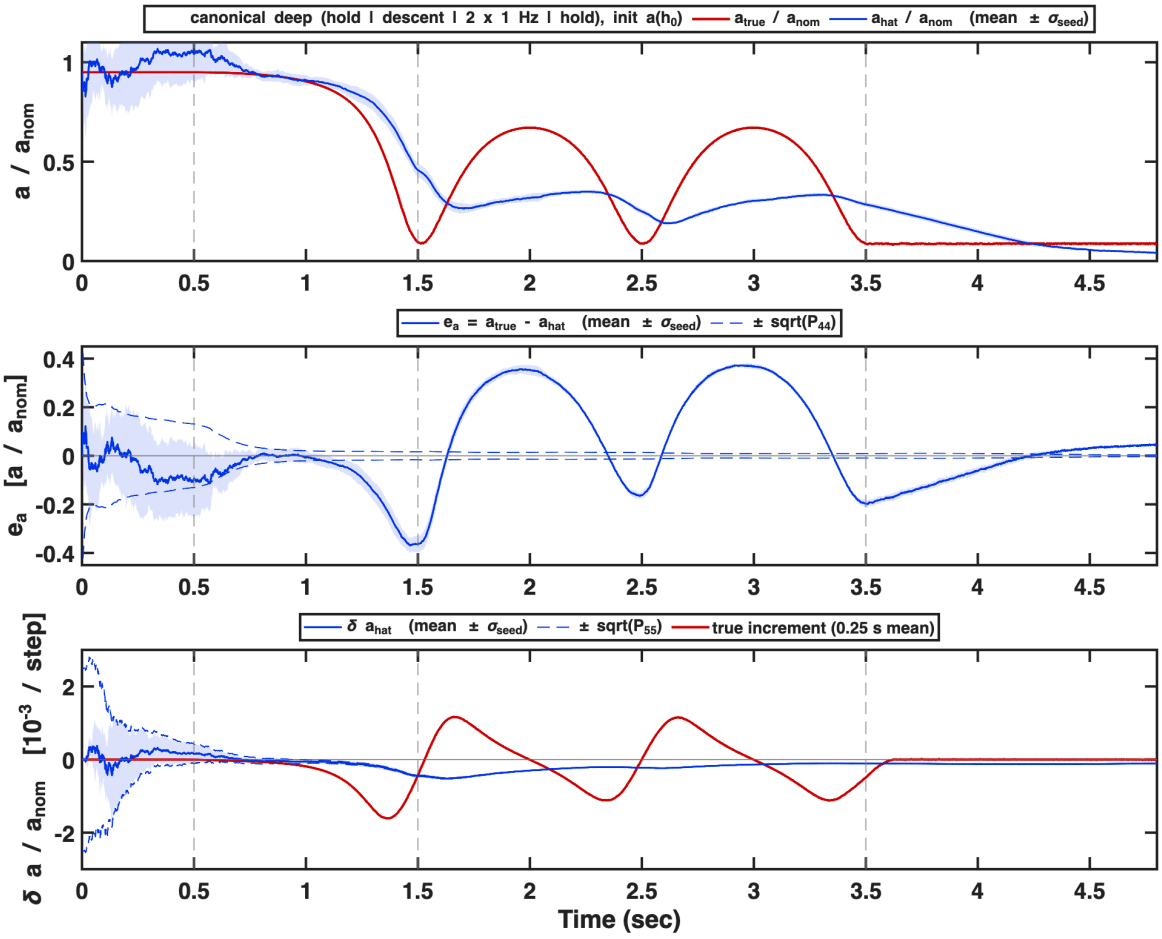


Fixed heights, far-field init $\hat{a}_x[0] = a_{nom}$: rows = gain, error with $\pm\sqrt{P_{44}}$, rate. Level converges at every height (last-second mean error $\leq 0.2\%\, a_{nom}$); the rate slot dies within 0.5 s.

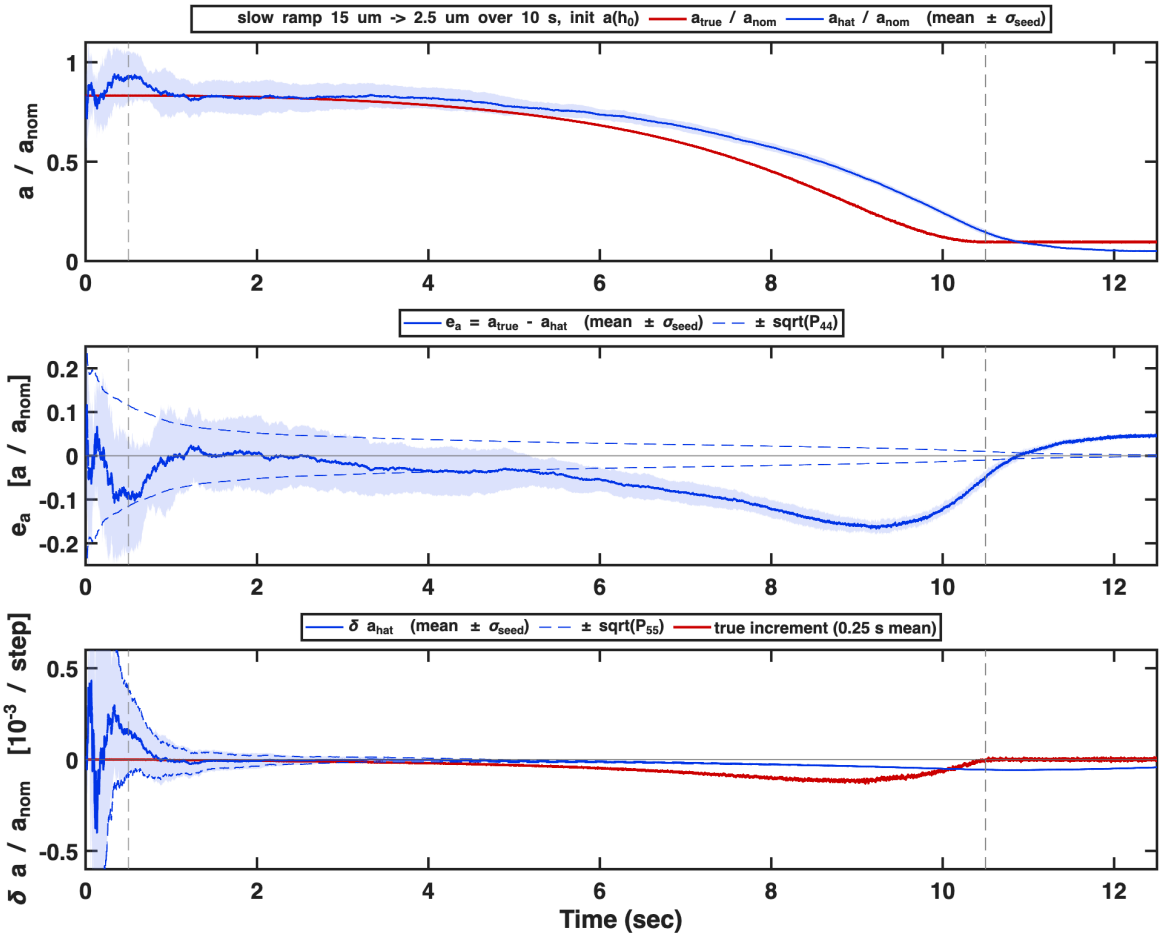
13. The two honesty gaps, discriminated (`probe_5state_noap_honesty.m`)

Gap 1, hold P 1.5–1.8× overconfident. Not the metric: against $a_{det}(\bar{h}_d)$ the ratio is unchanged (1.53 at 22.2 R where $a_{ram} \equiv 0$). Not the filter: with a white Gaussian oracle u of the same mean and marginal (IF = 1) the ratio is **0.96**. The measured colour of the real u ($\rho_u = 0.65, 0.36, 0.13, \ldots$, $1 + 2\sum\rho = 3.63$) matches the booked $\mathrm{IF}_{white} = 3.75$, so the DC factor is right and the residual is what "white with inflated R" cannot represent: a correlated sample stream feeding a two-slot (level + rate) filter. Next instrument, not tuning: an explicit MA/AR model of u in the measurement equation (Meng ledger 40 `y2_ma2` precedent).

Gap 2, y_1 1.76 in the oscillation. In 0.25 s bins the ratio is 2.0–2.8 exactly where $e_a \approx -0.3$ ($\hat{a}_x$ below the truth at the gain peaks) and 0.5–0.75 where $e_a > 0$; the bin values follow $a_x/\hat{a}_x$ (e.g. $0.62/0.29 = 2.1$ vs 2.35 measured), i.e. the thermal container $Q_T = 4k_BT\hat{a}_x$ evaluated at a gain that is wrong by 2–3×. Evaluating Q_T at a_x (`q_gain_true`) moves the window ratio 1.76 → 1.53 and makes descent / hold-end honest (1.04 / 1.05); the oracle y_2 changes nothing; the true-gain law does not remove it (2.09). The remaining 1.5 is [hypothesis] the deterministic $F_{dx}e_a$ swing counted as variance against a P_{44} that is 30× too small — the frozen-rate model's own error. Both gaps are consequences of $|e_a| \sim 0.3$, i.e. of §10 item 4, not separate defects.



Canonical deep trajectory, cheat init: the rate estimate (row 4, blue) stays at zero while the true increment swings $\pm 2 \times 10^{-3}$ per step; the level follows through the y_2 row only, as a low-pass of the swing (row 3, $\pm 0.35\, a_{nom}$, $\sqrt{P_{44}} = 0.01$).

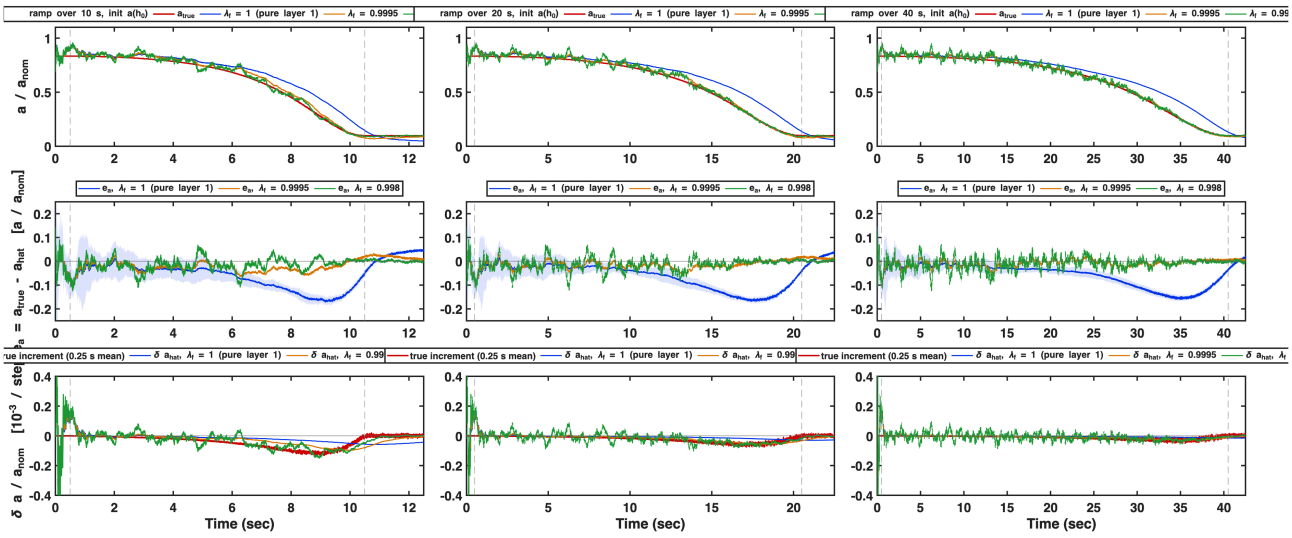


Slow ramp (15 to 2.5 μm over 10 s): lag builds to $+0.17\,a_{nom}$ at the ramp end and relaxes to -0.02 in the hold; the rate slot never carries the ramp.

14. Slow ramps and the forgetting factor (2026-09-21; `run_5state_noap_slow.m`, 5 seeds)

House ramp 15 → 2.5 μm over $T_d \in \{10, 20, 40\}$ s. Pure layer 1: the lag error peak is $-0.17/-0.17/-0.16\,a_{nom}$ at all three speeds, because the level channel's time constant grows with the record (τ = 0.90 / 1.74 / 3.14 s, $P_{44} \propto 1/N$): slowing the command does not help a filter with infinite memory, $\tau \cdot \dot{a}$ stays put. $\sqrt{P_{55}}$ at the fastest fall is 50–150× below the true increment.

Layer 1.5 = gain-block forgetting λ_f (slots 4–5 of P divided by λ_f before each predict; a memory knob, $T^* = T_s/(1-\lambda_f)$): 0.9995 (T^* = 1.25 s) gives lag peak $-0.065/-0.056/-0.035$, $\tau \approx 0.2$ s, rate slot alive (row 3), hold-end honesty 1.09 / 0.93 / 0.70; 0.998 (T^* = 0.31 s) buys no further lag reduction and doubles the hold-end scatter (0.009 vs 0.005). Tracking rms on the ramp 25 nm in every cell: the gain error never reaches the position (06-18). The first-second transient (±0.1, §13) is unchanged by λ_f.



Columns T_d = 10 / 20 / 40 s; rows gain, $e_a = a_{true} - \hat{a}$, rate. Blue = pure layer 1, orange $\lambda_f = 0.9995$, green 0.998.

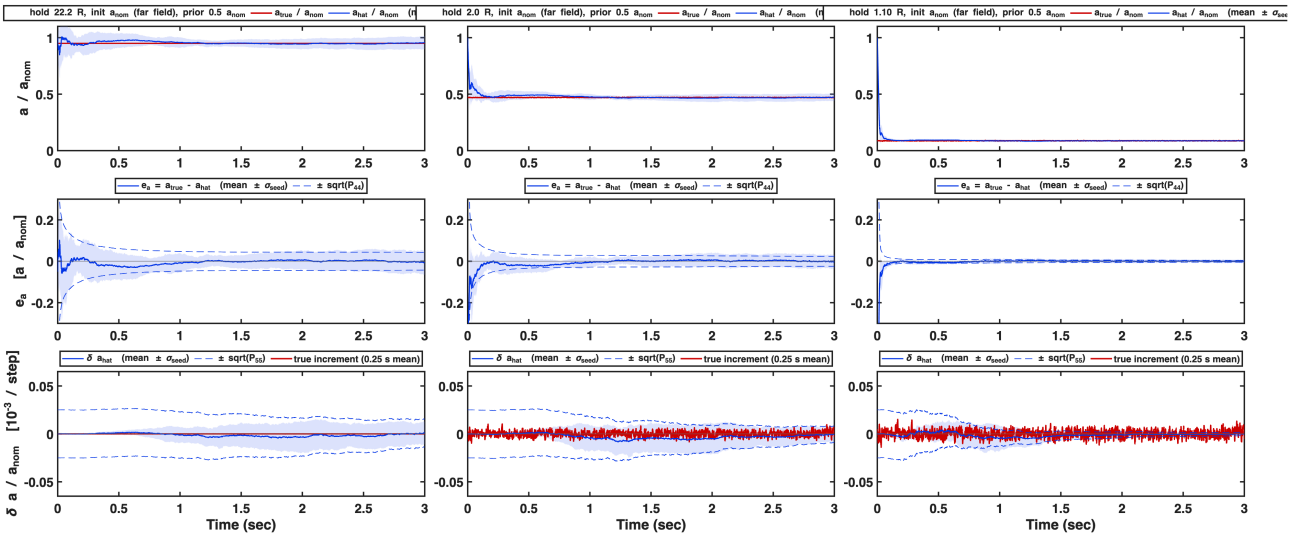
15. Closing state of the layer (2026-09-22)

Final defaults (all in the controller): gain model order 2 (level + rate), $Q_{44} = Q_{55} = 0$; memory schedule `lf_mode = 'ljung'` on the gain block (slots 4, 5, 8), clamp $\lambda_f \geq 0.98$ ($T^* \geq 31$ ms; the 1 Hz canon asks for ∼30 ms, the slow ramp pays 5.0 → 5.3% rel. rms); $\sqrt{P_{44}[0]} = 0.5\, a_{nom}$ (far-field seed) / $0.05\, a_{nom}$ (cheat seed); $\sqrt{P_{55}[0]} = a_{nom}/40000$ (runs start in a hold; $a_{nom}/400$ let the first correlated samples be read as a slope: start transient ±0.09 → ±0.02, ramp and hold unchanged).

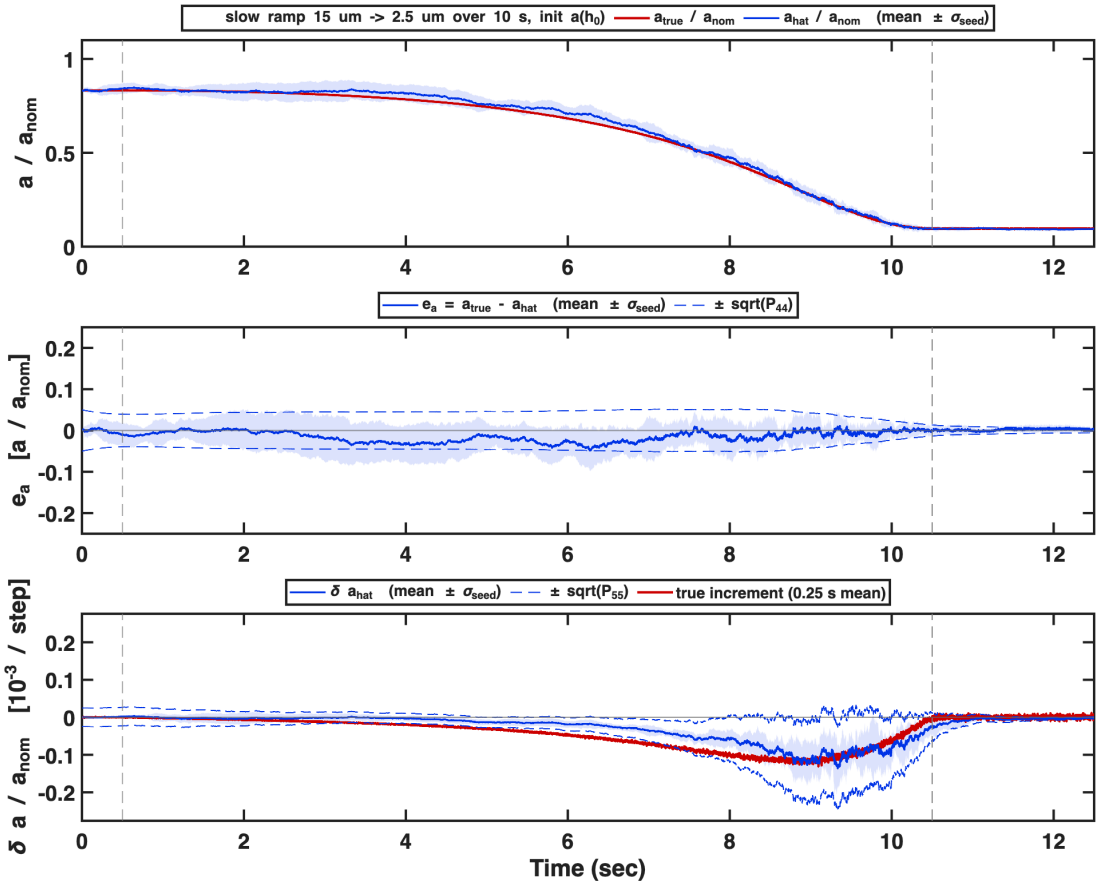
Five-cell battery, plane, 10 seeds (`run_5state_noap_battery` tag `_best`; figures below): fixed 22.2 / 2.0 / 1.10 R last-second mean ≤ 0.002, seed sd 0.051 / 0.033 / 0.008 vs $\sqrt{P_{44}}$ 0.043 / 0.025 / 0.005 (honesty 1.2 / 1.3 / 1.5); slow ramp descent −0.015±0.037 ($\sqrt{P_{44}}$ 0.044), hold-end −0.002±0.008 (1.02); canon oscillation $+0.05 \pm 0.05$ ($\sqrt{P_{44}}$ 0.07), hold-end -0.002 ± 0.010. Cell wall (plane slid 1.05 R): same absolute levels, relative 5× smaller (the estimator reads no wall).

Variants measured and retired (5 seeds unless noted): gain order 1 worst on both trajectories (rate slot needed); order 3 canon rms 0.066 → 0.041 at hold sd 0.009 → 0.013, no gain on the ramp (flag `gain_order`); damped rate (Singer, $\rho = 1 - 1/T^*$, $Q_{55} = (1-\rho^2)(r\hat{a})^2$) canon rms → 0.037, max 0.143 → 0.086, ramp +14% (flag `rate_damp`); forgetting on the rate slot only equal on the ramp, canon rms 0.105 and hold-end bias +0.04 (flag `lf_slots`); constant λ_f with rate-only forgetting: 0.9995 lags and holds +0.03, 0.998 far-field ±0.15; exact wall slope in r identical to the far-field slope (clamp binds); Q_{55} budget $(ra)^2/T^*$ equal to forgetting; gauge-sensitivity $H_2 \times s_i$ (`y2_mirror`) biases the 1.10 R hold by −0.01 and breaks the ramp (0.029 → 0.130): kept off.

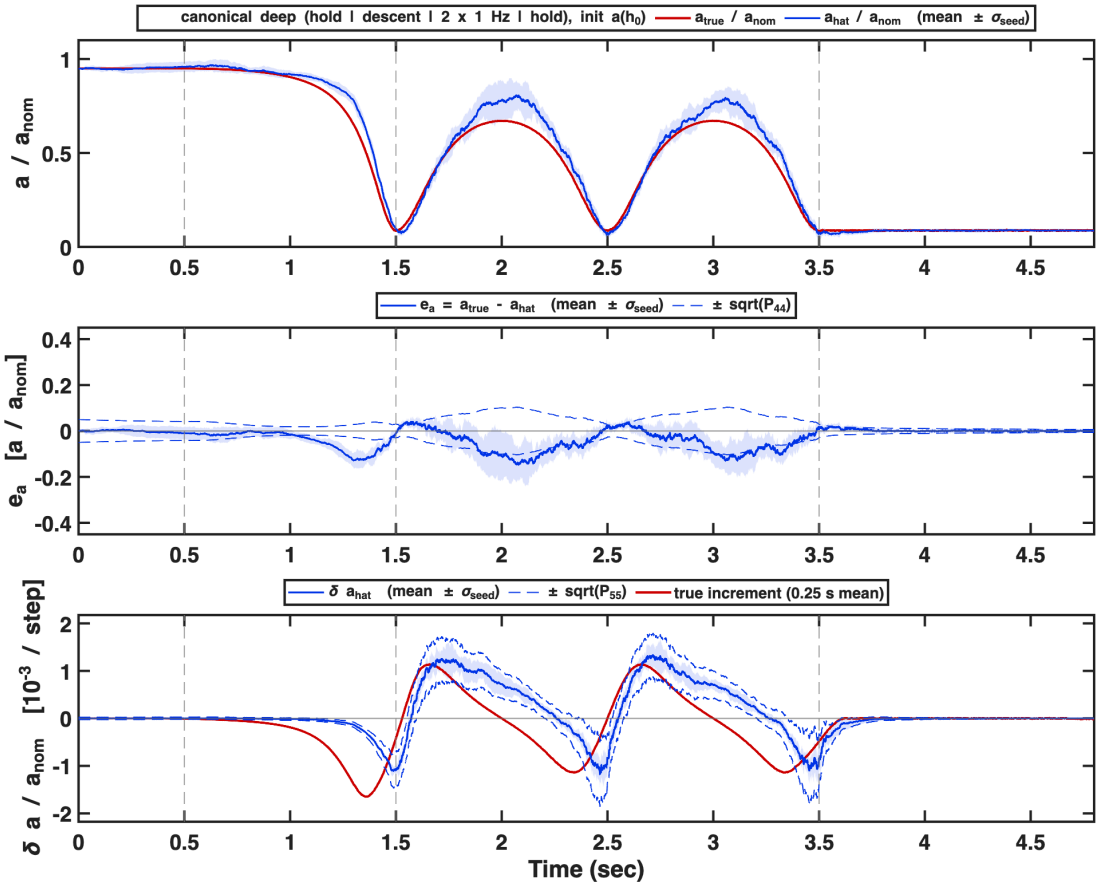
What the layer shows: at a fixed point the scatter plateaus at 5–9% of a after 0.5 s and does not fall further; on the slow ramp the scatter sits on the same plateau (the ramp is the fixed point at its 0.4–0.6 s averaging time) and the mean lag is removed by the schedule; on the 1 Hz canon the turning-point spike ∼ 0.09–0.15 is the memory limit on a reversing rate. Slowing the command buys speed$^{1/3}$. The plateau is not the loop echo (true-gain law: 1.52 → 1.41 at 1.10 R) and not the filter (white oracle y_2: 0.96); it is the readout stream's structure, left as the open item of the layer (a readout noise model, not built). Everything beyond this needs $a'\,\Delta\bar{w}_d$: the next rung.



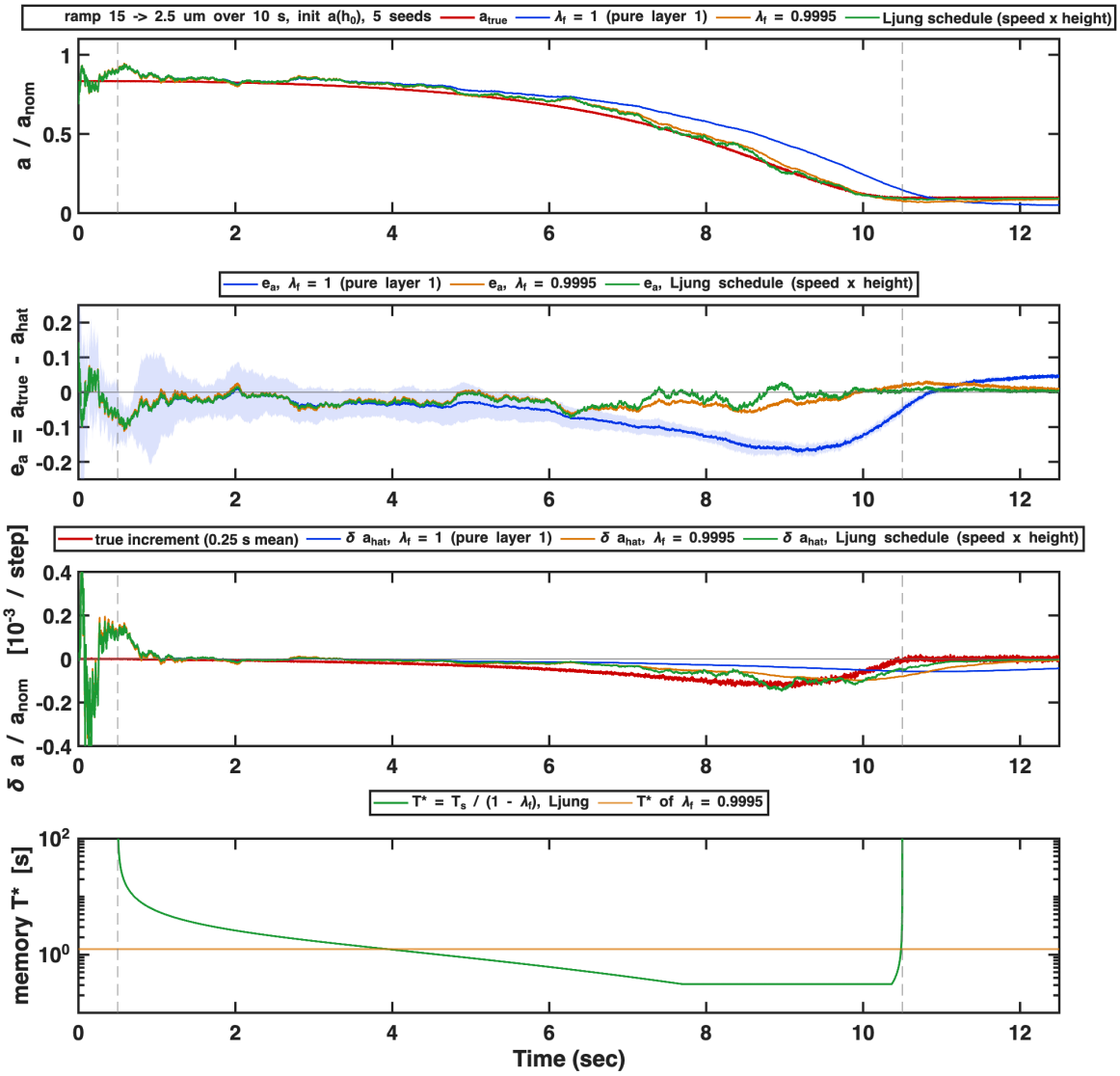
Final defaults, plane, fixed heights.



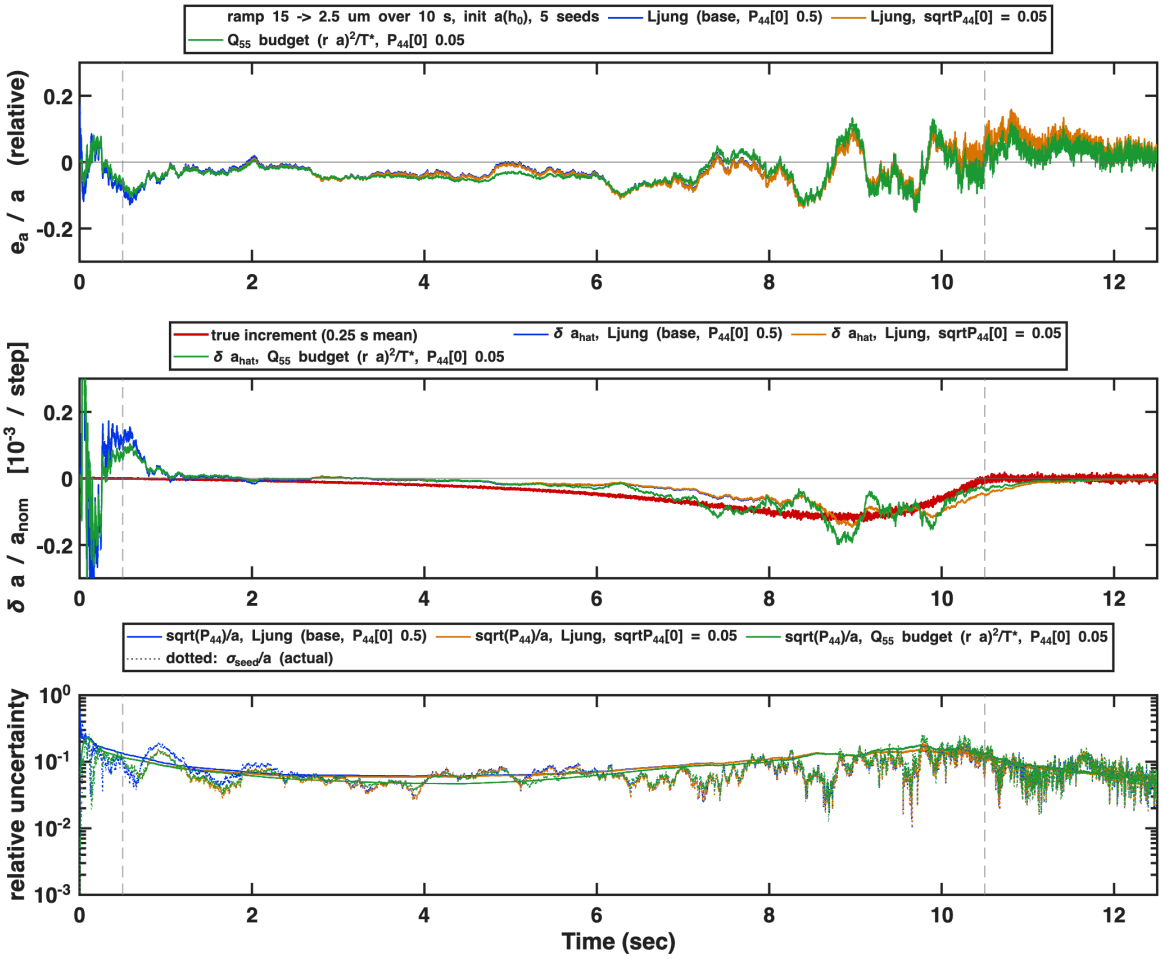
Final defaults, plane, slow ramp.



Final defaults, plane, canonical deep.



Slow ramp: pure layer 1 / constant λ_f / Ljung schedule (`run_5state_noap_lfadapt`).



Tuning round on the slow ramp (`run_5state_noap_tune`): $P_{44}[0]$, clamp, exact slope, Q_{55} budget.